

Scaling the HSF Conditions Database Across Experiments

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Conditions Data

Conditions data is any additional data needed to process event data

WHAT IS IT ?	EXAMPLES	VALIDITY RANGE
• Non-event parameters that change over time • Needed for calibration, alignment, and reconstruction	• Calibration constants, alignment geometry • Beam parameters, magnetic field maps • Bad channel maps	• Each conditions data file (payload) is valid for a specific range of runs • Ensures correct conditions are applied to each event

Key concept — Interval of Validity (IOV): Each conditions payload is valid for a specific range of runs/events. For example, a calorimeter calibration may be valid for **Run 1000 – Run 1500**, then **Run 1300 – Run 1500** replaced by an updated set.

Database architecture

Edit permissions for a Global Tag: unlocked, locked, or frozen

Global Tag Status



Category of conditions data (e.g. CalorimeterCalib")

Payload Type



Global Tag



Payload List



Payload IOV



Payload Data

Consistent, named snapshot of all conditions

Collection of IOVs for one data type within a Global Tag

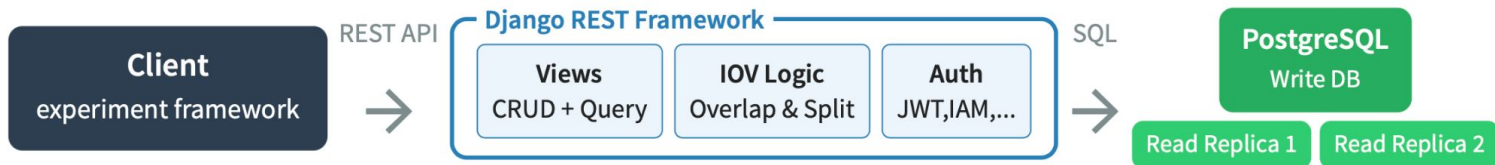
Conditions record with a validity range (start/end)

Actual file stored on external storage

- **Reproducibility** — “frozen” Global Tags guarantee identical conditions for reprocessing
- **Versioning** — clone & modify Global Tags for new processing campaigns
- **Non-overlapping IOVs** — exactly one valid payload per type at any point in time
- **Payload decoupling** — metadata (DB) is decoupled from payload files (storage)

Server implementation

REST API + PostgreSQL with optimized IOV lookups



- Development started in 2021
- Django-based server: REST API with auth/authz and experiment-specific plugins
- Web UI in planning

- C++ client library + CL tools; client handles payload download/upload
- Helm chart for one-shot deployment on OpenShift or any Kubernetes platform

OpenShift / Kubernetes — Helm charts



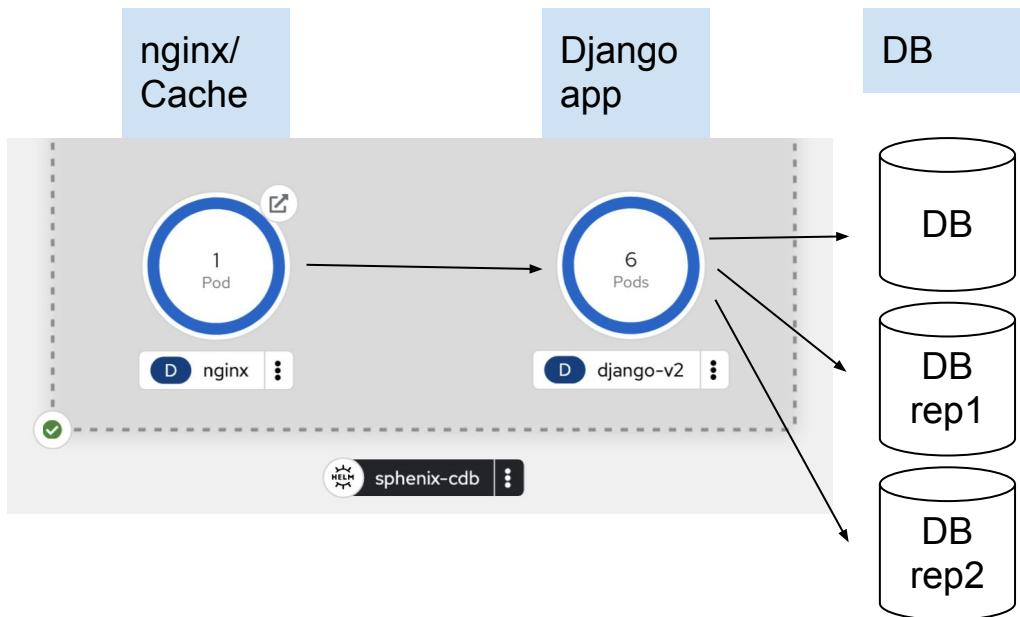
sPHENIX experiment on RHIC

sPHENIX Conditions Data Service

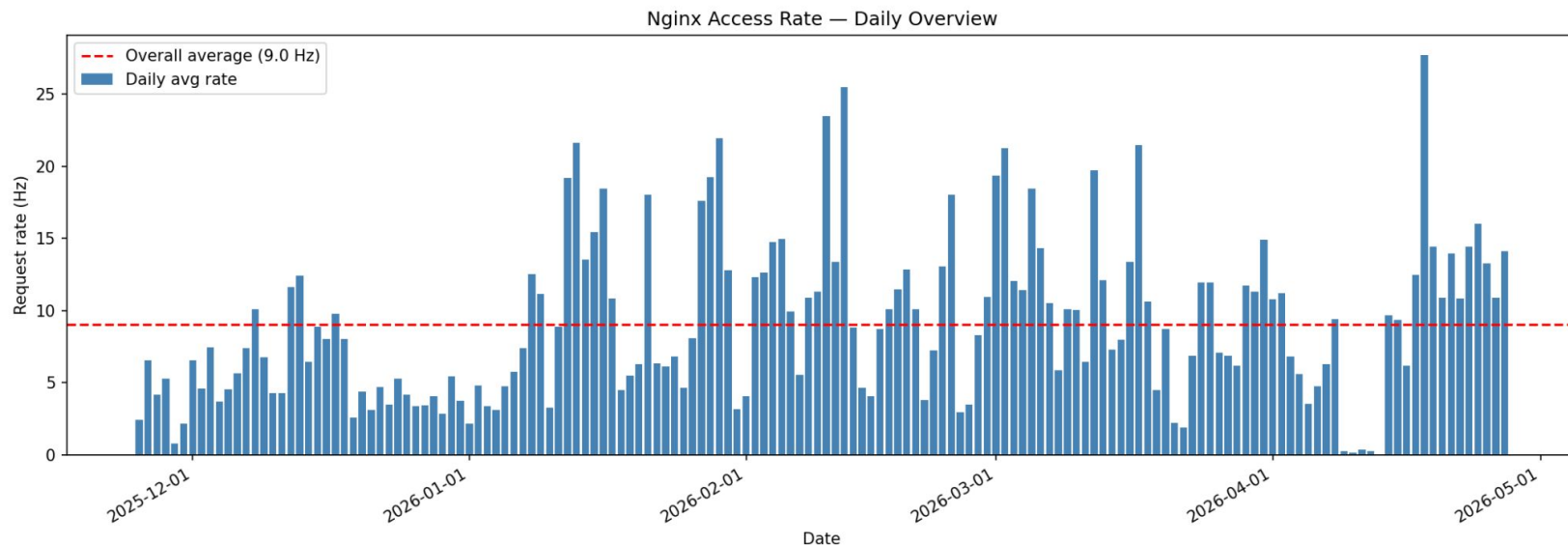
- Supports over 70,000 concurrent jobs on a farm running 132,000 logical cores
- Timeline
 - In production since January 2022
 - [HSF Conditions Database reference implementation](#) - CHEP2023
 - PSQL cluster provided for sPHENIX - December 2024
- Conditions data snapshot:
 - 133 global tags
 - Avg global tag size: 76 payload types, 19017 IOVs
 - Median global tag size: 76 payload types, 1881 IOVs
 - Global tag with most payload types: 163 types
 - Global tag with most IOVs: 133703 IOVs
 - In total, there are 204593 payload files totaling 26 GB in size, with a rough average of 127 KB/file

sPHENIX CDB instance

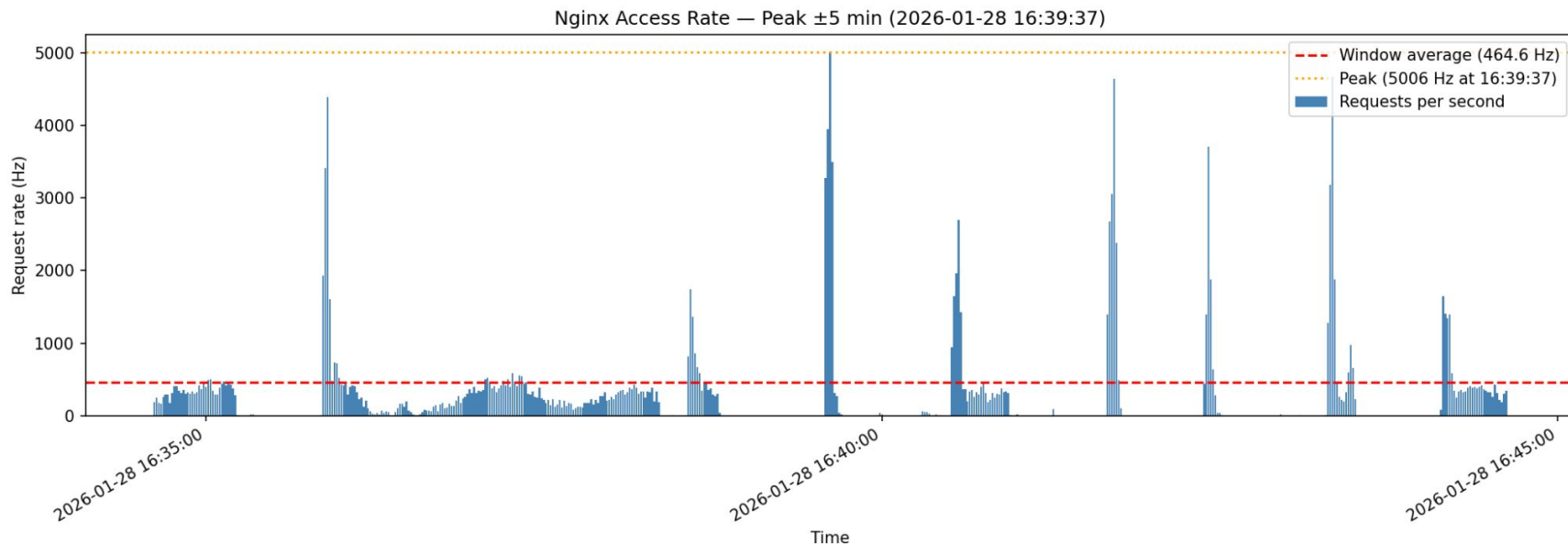
- Host: OpenShift@SCDF BNL
- Pods: 1 nginx, 6 Django - 4 threads each
- PostgreSQL
 - 3 VMs: 1 main, 2 replicas
 - CPU: 8 cores,
 - RAM: 32 GB
 - max_conn 5000
 - DB size: 3GB
- Payloads are handled by the client



sPHENIX conditions data daily average access rate



sPHENIX conditions data access pattern

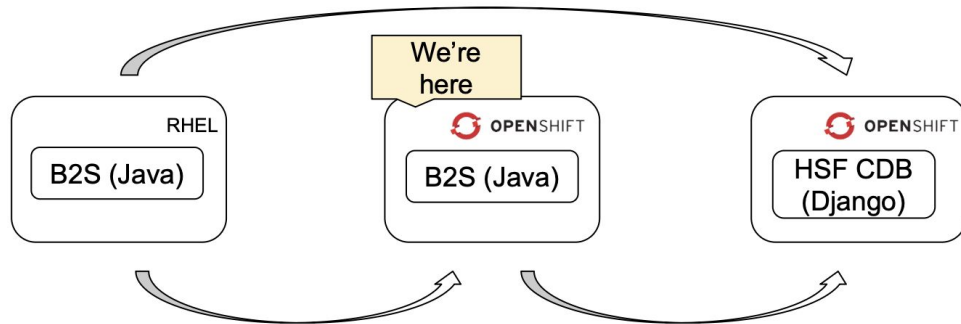


Belle II experiment on SuperKEKB

Belle II Conditions Data service

- Belle II runs up to about 36,000 concurrent jobs across about 38 grid sites
- Timeline
 - Belle II uses B2S - Java implementation of the CDB
 - HSF CDB migration has been announced in June 2024
 - Belle II migrated to OpenShift in June 2025
 - Belle II estimates final migration to the HSF CDB in 2027

The original [Java CDB \(B2S\)](#) and its VM-based deployment on RHEL could not sustain request rates beyond 100-200 Hz

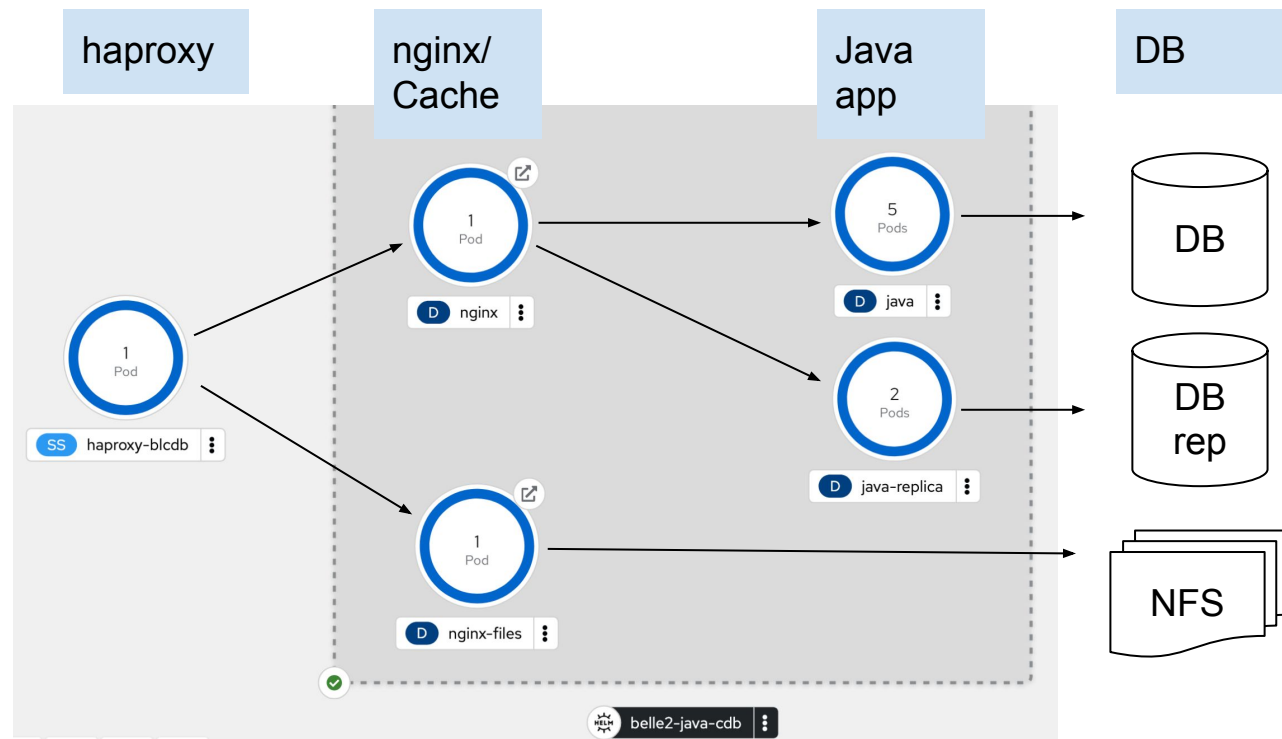


Belle II Conditions Data accounting

- Conditions data snapshot:
 - 3705 global tags
 - Avg global tag size: 86 payload types, 4273 IOVs
 - Median global tag size: 3 payload types, 12 IOVs
 - Global tag with most payload types: 2840 types
 - Global tag with most IOVs: 276124 IOVs
 - In total, there are 572048 payload files (in 10282 directories) totaling 27 GB in size, with a rough average of 47.2 KB/file

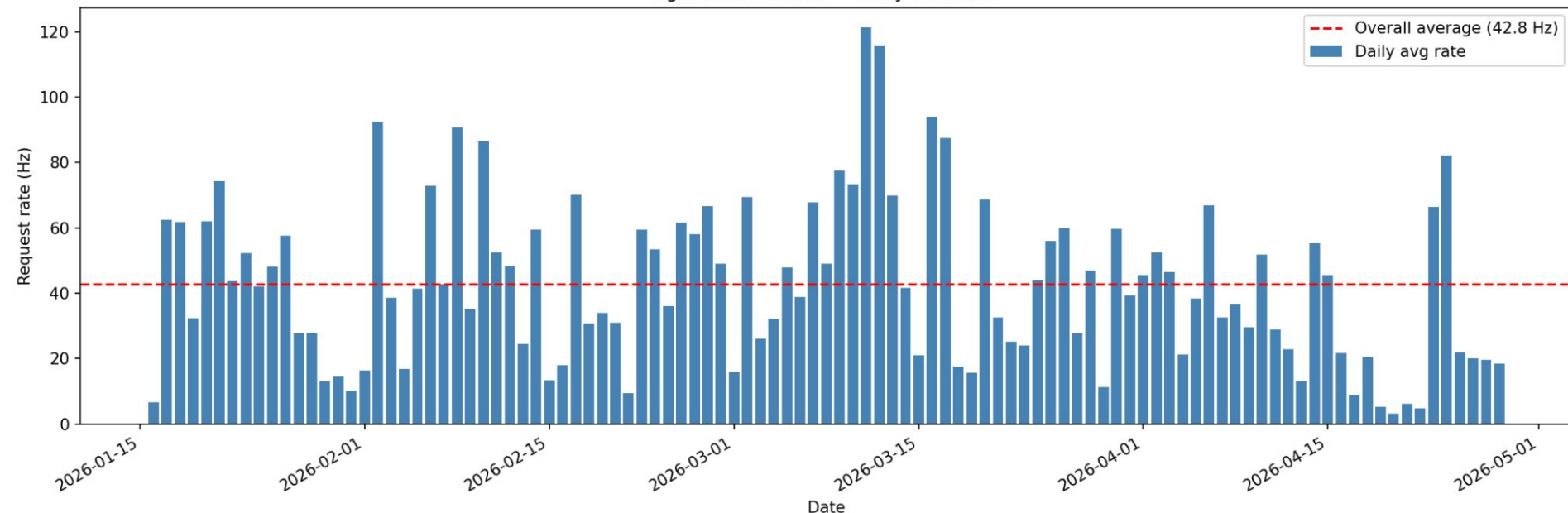
Belle II current Java instance

- Host:
OpenShift@SCDF BNL
- HAProxy pod exposes the instance externally
- 2 Nginx pods
 - Metadata Service
 - Http access to payloads
- 7 Java pods - 32+ threads
- PostgreSQL
 - VM in OpenShift
 - 32 CPU cores, 64G memory
- DB size 3.8 GB

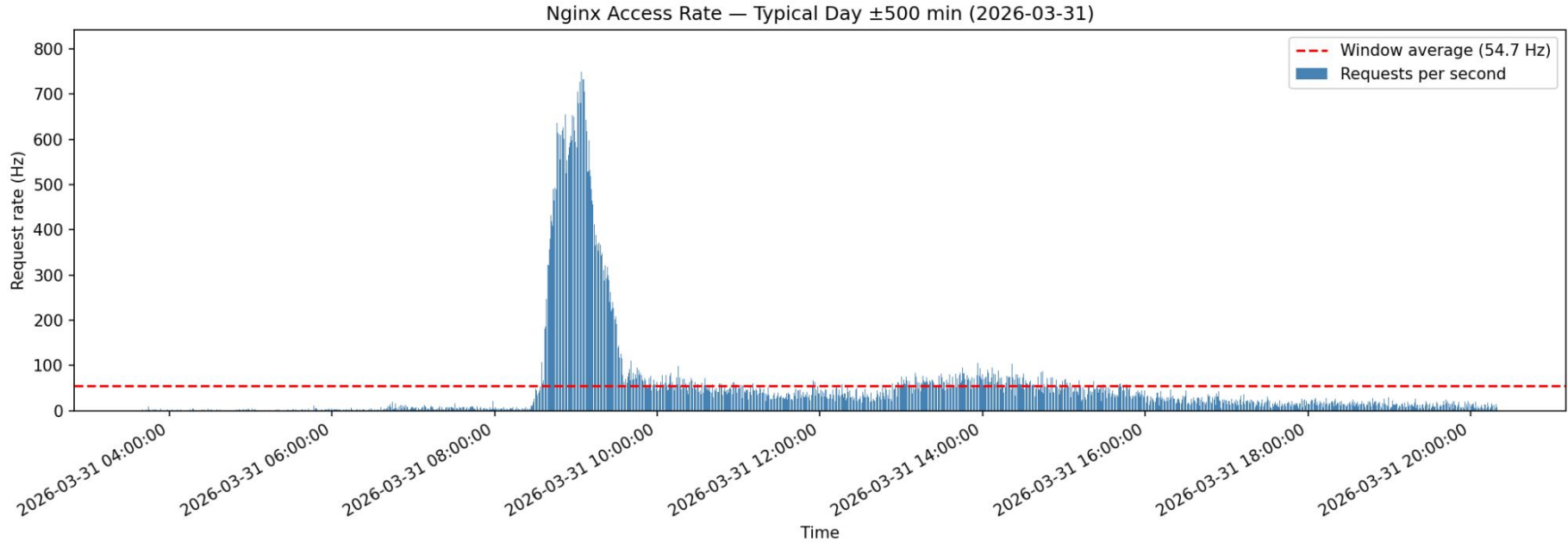


Belle II conditions data daily average access rate

Nginx Access Rate — Daily Overview

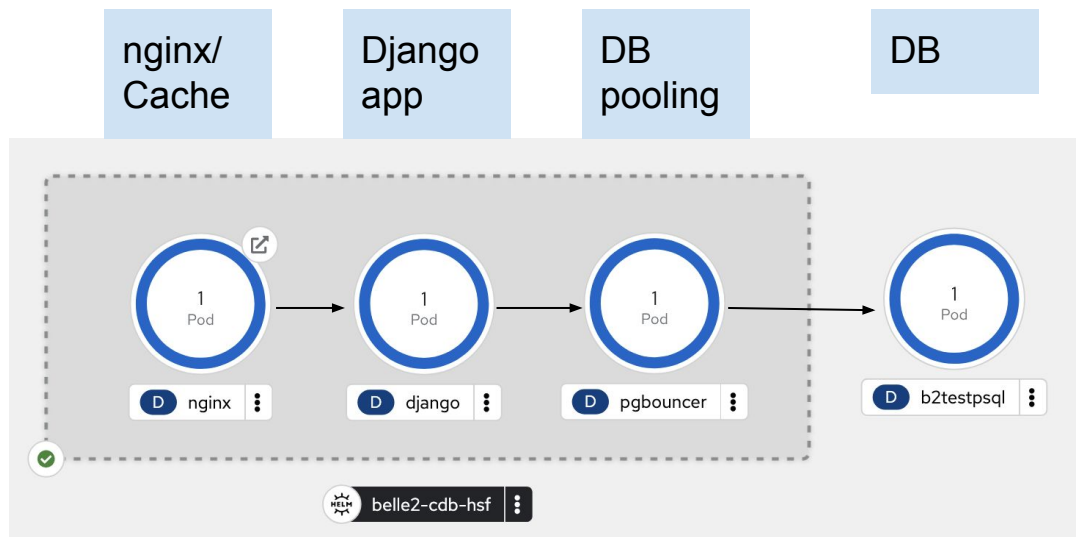


Belle II conditions data access pattern



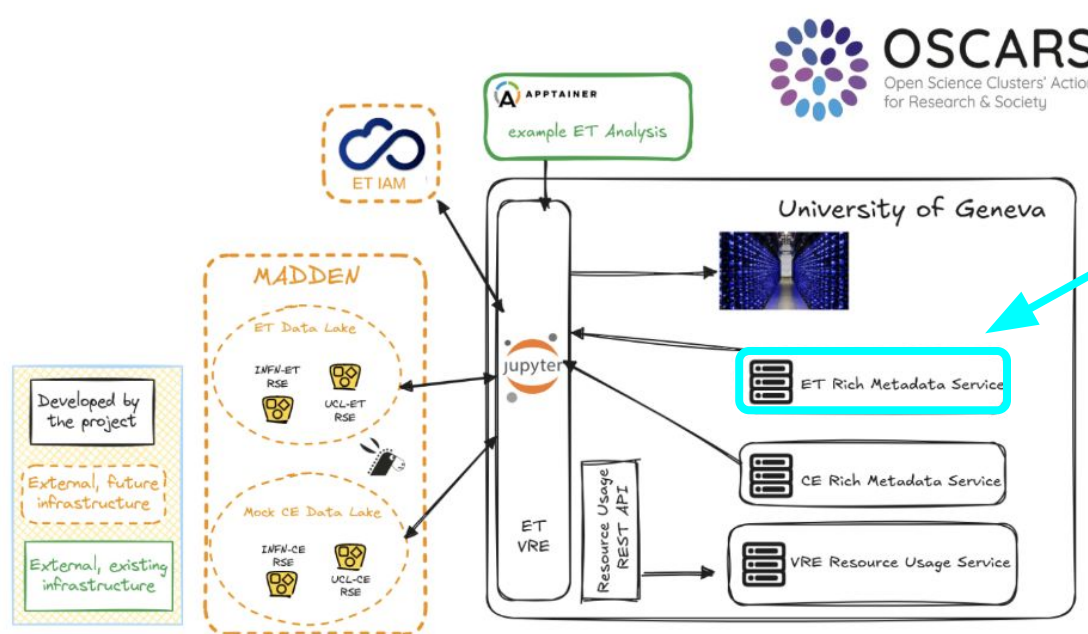
Belle2 HSF CDB test instance

- Test instance at OpenShift@SCDF BNL
- Minimal deployment - 4 pods
- Used for the conditions data migration and tests
- Will be gradually reconfigured for production



Einstein Telescope

Einstein Telescope Analysis Portal (EOSC and OSCARs)



OSCARS
Open Science Clusters' Action
for Research & Society

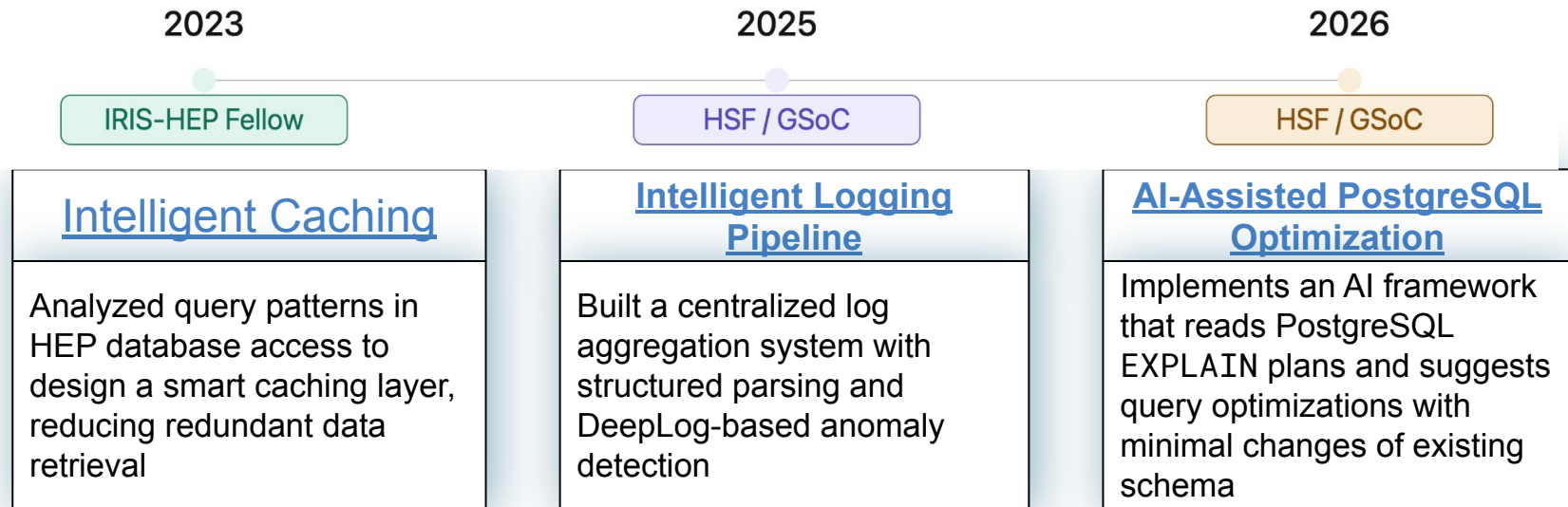
EOSC cascading grants (OSCARS) project - [ETAP](#)

Proposal to include the HSF CDB as a generic “rich metadata service”

Main work - repackage helm chart to include it in the ESCAPE VRE

Status - see [CHEP poster](#)

HSF/GSoC and IRIS-HEP projects



- **Applying data-driven and AI techniques** to make HEP computing infrastructure more efficient
- **Growing the next generation of HEP software engineers** through HSF and IRIS-HEP mentorship programs
- **Bridging physics experiments and modern software engineering**

Conclusion

- Status
 - Production since 2022 (sPHENIX): 70k+ concurrent jobs, 132k cores, peaks to 5000 Hz
 - Belle II migration underway — target 2027: about 36,000 concurrent jobs running across about 38 grid sites
 - Being evaluated by Einstein Telescope / ETAP
- Development
 - Web UI development
 - Kubernetes-native PostgreSQL cluster (GSoC 2026)
 - Anomaly detection and intelligent log analysis (GSoC 2025)

Thank you

Backup

Chronicles

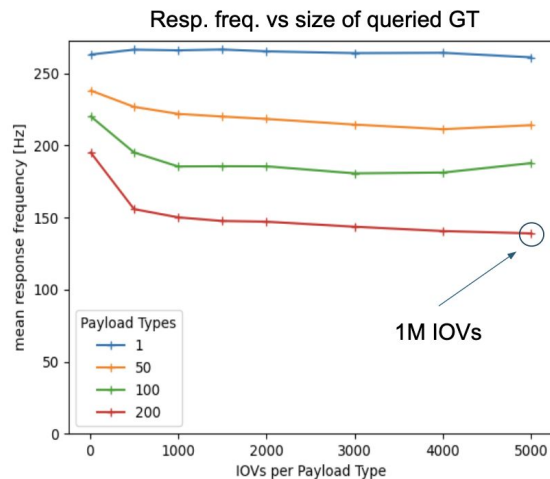
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Initial performance testing

- Initial tests on OKD
 - Co-located client
 - Single-pod DB (with pooling)
 - No cache

sPHENIX-driven scenarios

Scenario	Payload Types	Payload IOVs (per type)
tiny	10	100 (10)
tiny-moderate	10	2000 (200)
moderate	100	20000 (200)
heavy-usage	100	500000 (5000)
worst-case	200	5200000 (26000)



Achieved mean values:

- 40 ms response time
- 230 Hz response frequency

